

CA-C30L: Mobile Application Development Lab

LAB PROGRAM-4 : EXPLICIT-IMPLICIT INTENT-CONTENT PROVIDER

4. Create an application to implement new activity using explicit intent, implicit intent and content provider.

Program description :

This is an improved version of previous program, where new activity is created and then content provider is added.

Content provider helps the applications to access the content from other applications .

In this program we are accessing the content from contacts of our mobile .

At First , when the application runs , clicking on content providers will ask for sign in , once this is done , we can view our contacts.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/
android"
xmlns:app="http://schemas.android.com/apk/res-
auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:orientation="vertical"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
```

```
<Button
    android:id="@+id/btnExplicit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Explicit"
    android:layout_gravity="center"
    android:layout_marginTop="100dp"/>
```

```
<Button
    android:id="@+id/btnImplicit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Implicit"
    android:layout_gravity="center"
    android:layout_marginTop="100dp"
/>
```

```
<Button
    android:id="@+id/btnContacts"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:layout_marginTop="100dp"
    android:text="Content provider" />
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.contentprovider;
```

```
import ...
```

```
@Override
protected void onCreate(Bundle
```

```
savedInstanceState) {  
    super.onCreate(savedInstanceState);  
  
    setContentView(R.layout.activity_main);  
  
    btnExplicit = findViewById(R.id.btnExplicit);  
    btnImplicit = findViewById(R.id.btnImplicit);  
    btnContacts = findViewById(R.id.btnContacts);  
  
    // Explicit Intent  
    btnExplicit.setOnClickListener(v -> {  
        Intent intent = new  
Intent(MainActivity.this, MainActivity2.class);  
        intent.putExtra("message", "Hello from  
MainActivity");  
        startActivity(intent);  
    });  
  
    // Implicit Intent  
    btnImplicit.setOnClickListener(v -> {  
        Intent intent = new  
Intent(Intent.ACTION_VIEW);  
  
intent.setData(Uri.parse("https://www.google.com"));  
        startActivity(intent);  
    });  
  
    // Content Provider (Contacts)  
    btnContacts.setOnClickListener(v -> {  
        Intent intent = new  
Intent(Intent.ACTION_VIEW,  
ContactsContract.Contacts.CONTENT_URI);  
        startActivity(intent);  
    });  
}  
}  
});
```

- Right click on java folder select **New->activity->empty views activity**

MainActivity2.java and its corresponding xml file will be generated.

MainActivity2.java

- No changes in second activity java file

activity_main2.xml

Include one text view and display a text “ Hello I have been explicitly called .
Everything else remains the same .

```
<TextView
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Hello,I have been explicitly called"
```

```
></TextView>
```
